The Geometry of Truth: Emergent Linear Structure in LLM Representations of True/False Datasets

Samuel Marks
Northeastern University
s.marks@northeastern.edu

Max Tegmark
MIT

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Abstract

Large Language Models (LLMs) have impressive capabilities, but are prone to outputting falsehoods. Recent work has developed techniques for inferring whether a LLM is telling the truth by training probes on the LLM's internal activations. However, this line of work is controversial, with some authors pointing out failures of these probes to generalize in basic ways, among other conceptual issues. In this work, we use high-quality datasets of simple true/false statements to study in detail the structure of LLM representations of truth, drawing on three lines of evidence: 1. Visualizations of LLM true/false statement representations, which reveal clear linear structure. 2. Transfer experiments in which probes trained on one dataset generalize to different datasets. 3. Causal evidence obtained by surgically intervening in a LLM's forward pass, causing it to treat false statements as true and vice versa. Overall, we present evidence that at sufficient scale, LLMs *linearly represent* the truth or falsehood of factual statements. We also show that simple difference-in-mean probes generalize as well as other probing techniques while identifying directions which are more causally implicated in model outputs.

1 Introduction

Despite their impressive capabilities, large language models (LLMs) do not always output true text (Lin et al., 2022; Steinhardt, 2023; Park et al., 2023). In some cases, this is because they do not know better. In other cases, LLMs apparently know that statements are false but generate them anyway. For instance, Perez et al. (2022) demonstrate that LLM assistants output more falsehoods when prompted with the biography of a less-educated user. More starkly, OpenAI (2023) documents a case where a GPT-4-based agent gained a person's help in solving a CAPTCHA by lying about being a vision-impaired human. "I should not reveal that I am a robot," the agent wrote in an internal chain-of-thought scratchpad, "I should make up an excuse for why I cannot solve CAPTCHAs."

We would like techniques which, given a language model M and a statement s, determine whether M believes s to be true (Christiano et al., 2021). One approach to this problem relies on inspecting model outputs; for instance, the internal chain-of-thought in the above example provides evidence that the model understood it was generating a falsehood. An alternative class of approaches instead leverages access to M's internal state when processing s. There has been considerable recent work on this class of approaches: Azaria & Mitchell (2023), Li et al. (2023b), and Burns et al. (2023) all train probes for classifying truthfulness based on a LLM's internal activations. In fact, the probes of Li et al. (2023b) and Burns et al. (2023) are *linear probes*, suggesting the presence of a "truth direction" in model internals.

However, the efficacy and interpretation of these results are controversial. For instance, Levinstein & Herrmann (2023) note that the probes of Azaria & Mitchell (2023) fail to generalize in basic ways, such as to statements containing the word "not." The probes of Burns et al. (2023) have similar generalization issues, especially when using representations from autoregressive transformers. This suggests these probes may be identifying not truth, but other features that correlate with truth on their training data.

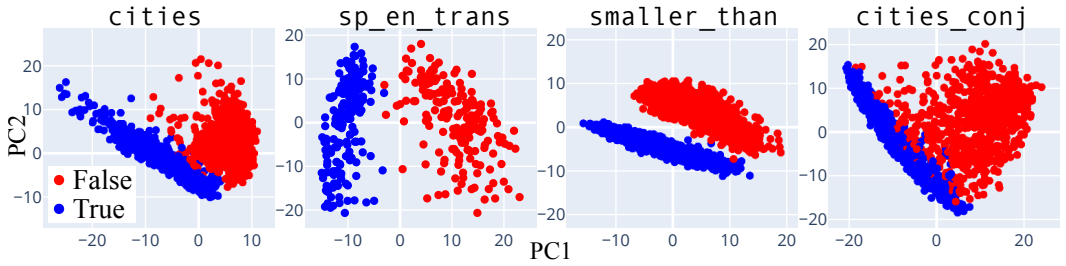

Figure 1: PCA visualizations for LLaMA-2-70B representations of our true/false datasets.

Working with autoregressive transformers from the LLaMA-2 family (Touvron et al., 2023), we shed light on this murky state of affairs. After curating high-quality datasets of simple, unambiguous true/false statements, we perform a detailed investigation of LLM representations of factuality. Our analysis, which draws on patching experiments, simple visualizations with principal component analysis (PCA), a study of probe generalization, and causal interventions, finds:

- **Evidence that linear representations of truth emerge with scale**, with larger models having a more abstract notion of truth that applies across structurally and topically diverse inputs.
- **A small group of causally-implicated hidden states** which encode these truth representations.
- Consistent results across a suite of probing techniques, but with **simple difference-in-mean probes identifying directions which are most causally implicated**.

Our code, datasets, and an interactive dataexplorer are available at `https://github.com/saprmarks/geometry-of-truth`.

1.1 Related work

Linear world models. Substantial previous work has studied whether LLMs encode world models in their representations (Li et al., 2023a; 2021; Abdou et al., 2021; Patel & Pavlick, 2022). Early work focused on whether individual neurons represent features (Wang et al., 2022; Sajjad et al., 2022; Bau et al., 2020), but features may more generally be represented by *directions* in a LLM's latent space (i.e. linear combinations of neurons) (Dalvi et al., 2018; Gurnee et al., 2023; Cunningham et al., 2023; Elhage et al., 2022). We say such features are *linearly represented* by the LLM. Just as other authors have asked whether models have directions representing the concepts of "West Africa" (Goh et al., 2021) or "basketball" (Gurnee et al., 2023), we ask here whether there is a direction corresponding to the truth or falsehood of a factual statement.

Probing for truthfulness. Others have trained probes to classify truthfulness from LLM activations, using both logistic regression (Azaria & Mitchell, 2023; Li et al., 2023b), unsupervised (Burns et al., 2023), and contrastive (Zou et al., 2023; Rimsky et al., 2024) techniques. This work differs from prior work in a number of ways. First, a cornerstone of our analysis is evaluating whether probes trained on one dataset transfer to topically and structurally different datasets in terms of *both* classification accuracy *and* causal mediation of model outputs. Second, we specifically interrogate whether our probes attend to *truth*, rather than merely features which correlate with truth (e.g. probable vs. improbable text). Third, we localize truth representations to a small number of hidden states above certain tokens. Fourth, we go beyond the mass-mean shift interventions of Li et al. (2023b) by systematically studying the properties of difference-in-mean. Finally, we carefully scope our setting, using only datasets of clear, simple, and unambiguous factual statements, rather than statements which are complicated and structured (Burns et al., 2023), confusing (Azaria & Mitchell, 2023; Levinstein & Herrmann, 2023), or intentionally misleading (Li et al., 2023b; Lin et al., 2022).

Table 1: Our datasets

Name	Description	Rows
cities	"The city of [city] is in [country]."	1496
neg_cities	Negations of statements in cities with "not"	1496
sp_en_trans	"The Spanish word '[word]' means '[English word]'."	354
neg_sp_en_trans	Negations of statements in sp_en_trans with "not"	354
larger_than	"x is larger than y."	1980
smaller_than	"x is smaller than y."	1980
cities_cities_conj	Conjunctions of two statements in cities with "and"	1500
cities_cities_disj	Disjunctions of two statements in cities with "or"	1500
companies_true_false	Claims about companies; from Azaria & Mitchell (2023)	1200
common_claim_true_false	Various claims; from Casper et al. (2023)	4450
counterfact_true_false	Various factual recall claims; from Meng et al. (2022)	31960
likely	Nonfactual text with likely or unlikely final tokens	10000

2 Datasets

Curated datasets. Unlike some prior work (Lin et al., 2022; Onoe et al., 2021, *inter alia*) on language model truthfulness, our primary goal is *not* to measure LLMs' capabilities for classifying the factuality of challenging data. Rather, our goal is to understand: Do LLMs have a unified representation of truth that spans structurally and topically diverse data? We therefore construct **curated** datasets with the following properties:

1. **Clear scope**. We scope "truth" to mean factuality, i.e. the truth or falsehood of a factual statement. App. A further clarifies this definition and contrasts it with related but distinct notions, such as correct question-answering or compliant instruction-following.
2. **Statements are simple, uncontroversial, and unambiguous.** In order to separate our interpretability analysis from questions of LLM capabilities, we work only with statements whose factuality our models are very likely to understand. For example "Sixty-one is larger than seventy-four" (false) or "The Spanish word 'nariz' does not mean 'giraffe' " (true).
3. **Controllable structural and topical diversity.** We structure our data as a union of smaller datasets. In each individual dataset, statements follow a fixed template and topic. However, the inter-dataset variation is large: in addition to covering different topics, we also—following Levinstein & Herrmann (2023)—introduce structural diversity by negating statements with "not" or taking logical conjunctions/disjunctions (e.g. "It is the case both that s1 and that s2").

Uncurated datasets. In order to validate that the truth representations we identify also generalize to other factual statements, we use **uncurated** datasets adapted from prior work. These more challenging test sets consist of statements which are more diverse, but also sometimes ambiguous, malformed, controversial, or difficult to understand.

likely dataset. To ensure that our truth representations do not merely reflect a representation of probable vs. improbable text, we introduce a **likely** dataset, consisting of *nonfactual text* where the final token is either the most or 100th most likely completion according to LLaMA-13B.

Our curated, uncurated, and likely datasets are shown in Tab. 1; addition information about their construction is in App. H.

We note that for some of our datasets, there is a strong *anti*-correlation between text being probable and text being true. For instance, for neg_cities and neg_sp_en_trans, the truth value of a statement and the log probability LLaMA-2-70B assigns to it correlate at $r = -.63$ and $r = -.89$, respectively.[1] This is intuitive: when prompted with "The city of Paris is not in", LLaMA-2-70B judges "France" to be the most probable continuation (among countries),

[1]In contrast, the correlation is strong and positive for cities ($r = .85$) and sp_en_trans ($r = .95$).

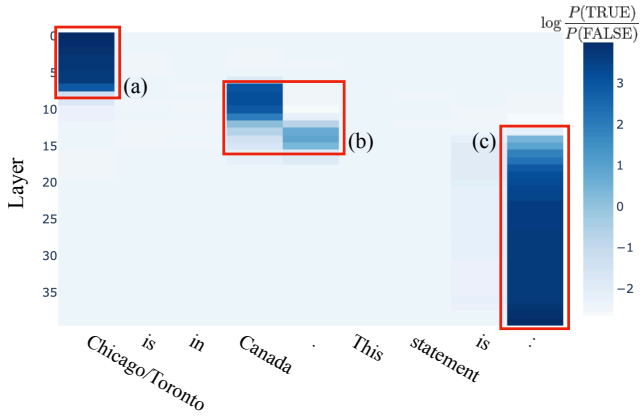

Figure 2: Difference $\log P(\texttt{TRUE}) - \log P(\texttt{FALSE})$ in LLaMA-2-13B log probabilities after patching residual stream activation in the indicated token position and layer.

despite this continuation being false. Together with the `likely` dataset, this will help us establish that the linear structure we observe in LLM representations is not due to LLMs linearly representing the difference between probable and improbable text.

3 Localizing truth representations via patching

Before beginning our study of LLM truth representations, we first address the question of which hidden states might contain such representations. We use simple patching experiments (Vig et al., 2020; Finlayson et al., 2021; Meng et al., 2022; Geiger et al., 2020) to localize certain hidden states for further analysis. Consider the following prompt p_F:

> The city of Tokyo is in Japan. This statement is: TRUE
> The city of Hanoi is in Poland. This statement is: FALSE
> The city of Chicago is in Canada. This statement is:

Similarly, let p_T be the prompt obtained from p_F by replacing "Chicago" with "Toronto," thereby making the final statement true. In order to localize causally implicated hidden states, we run our model M on the input p_T and cache the residual stream activations $h_{i,\ell}(p_T)$ for each token position i and layer ℓ. Then, for each i and ℓ, we run M on p_F but modify M's forward pass by swapping out the residual stream activation $h_{i,\ell}(p_F)$ for $h_{i,\ell}(p_T)$ (and allowing this change to affect downstream computations); for each of these intervention experiments, we record the difference in log probability between the tokens "TRUE" and "FALSE"; the larger this difference, the more causally influential the hidden state in position i and layer ℓ is on the model's prediction.

Results for LLaMA-2-13B and the `cities` dataset are shown in Fig. 2; see App. B for results on more models and datasets. We see three groups of causally implicated hidden states. The final group, labeled (c), directly encodes the model's prediction: after applying the LLM's decoder head directly to these hidden states, the top logits belong to tokens like "true," "True," and "TRUE." The first group, labeled (a), likely encodes the LLM's representation of "Chicago" or "Toronto."

What does group (b) encode? The position of this group—over the final token of the statement and end-of-sentence punctuation[2]—suggests that it encodes information pertaining to the full statement. Since the information encoded is also causally influential on the model's decision to output "TRUE" or "FALSE," we hypothesize that these hidden states store a

[2]This *summarization* behavior, in which information about clauses is encoded over clause-ending punctuation tokens, was also noted in Tigges et al. (2023). We note that the largest LLaMA model displays this summarization behavior in a more context-dependent way; see App. B.

representation of the statement's truth. In the remainder of this paper, we systematically study these hidden states.

4 Visualizing LLM representations of true/false datasets

We begin our investigation with a simple technique: visualizing LLMs representations of our datasets using principal component analysis (PCA). Guided by the results of §3, we present here visualizations of the *most downstream* hidden state in group (b); for example, for LLaMA-2-13B, we use the layer 15 residual stream activation over the end-of-sentence punctuation token.[3] Unlike in §3, we do not prepend the statements with a few-shot prompt (so our models are not "primed" to consider the truth value of our statements). For each dataset, we also center the activations by subtracting off their mean.

When visualizing LLaMA-2-13B and 70B representations of our curated datasets – datasets constructed to have little variation with respect to non-truth features, such as sentence structure or subject matter – **we see clear linear structure** (Fig. 1), with true statements separating from false ones in the top two principal components (PCs). As explored in App. C, this structure emerges rapidly in early-middle layers and emerges later for datasets of more structurally complex statements (e.g. conjunctive statements).

To what extent does this visually-apparent linear structure align between different datasets? Our visualizations indicate a nuanced answer: **the axes of separation for various true/false datasets align often, but not always**. For instance, Fig. 3(a) shows the first PC of `cities` also separating true/false statements from other datasets, including diverse uncurated datasets. On the other hand, Fig. 3(c) shows stark failures of alignment, with the axes of separation for datasets and statements and their negations being approximately orthogonal.

These cases of misalignment have an interesting relationship to scale. Fig. 3(b) shows `larger_than` and `smaller_than` separating along *antipodal* directions in LLaMA-2-13B, but along a common direction in LLaMA-2-70B. App. C depicts a similar phenomenon occuring over the layers of LLaMA-2-13B: in early layers, `cities` and `neg_cities` separate antipodally, before rotating to lie orthogonally (as in Fig. 3(c)), and finally aligning in later layers.

4.1 Discussion

Overall, these visualizations suggest that as LLMs scale (and perhaps, also as a fixed LLM progresses through its forward pass), they hierarchically develop and linearly represent increasingly general abstractions. Small models represent surface-level characteristics of their inputs, and large models linearly represent more abstract concepts, potentially including notions like "truth" that capture shared properties of topically and structurally diverse inputs. In middle regimes, we may find linear representation of concepts at intermediate levels of abstraction, for example, "accurate factual recall" or "close association" (in the sense that "Beijing" and "China" are closely associated).

To explore these intermediate regimes more deeply, suppose that $\mathcal{D}$ and $\mathcal{D}'$ are true/false datasets, f^+ is a linearly-represented feature which correlates with truth on both $\mathcal{D}$ and $\mathcal{D}'$, and f^- is a feature which correlates with truth on $\mathcal{D}$ but has a negative correlation with truth on $\mathcal{D}'$. If f^+ is very *salient* (i.e. the datasets' have large variance along the f-direction) and f^- is not, then we expect PCA visualizations of $\mathcal{D} \cup \mathcal{D}'$ to show joint separation along f^+. If f^- is very salient but f^+ is not, we expect antipodal separation along f^-, as in Fig. 3(b, center). And if both f^+ and f^- are salient, we expect visualizations like Fig. 3(c).

To give an example, suppose that $\mathcal{D} =$ `cities`, $\mathcal{D}' =$ `neg_cities`, $f^+ =$ "truth", and $f^- =$ "close association". Then we might expect f^- to correlate with truth positively on $\mathcal{D}$ and negatively on $\mathcal{D}'$. If so, we would expect training linear probes on $\mathcal{D} \cup \mathcal{D}'$ to result in

[3] Our qualitative results are insensitive to choice of layer among early-middle to late-middle layers. On the other hand, when using representations over the final token in the statement (instead of the punctuation token), we sometimes see that the top PCs instead capture variation in the token itself (e.g. clusters for statements ending in "China" regardless of their truth value).